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RNA bioinformatics: the basics behind the 16 projects

What RNA structure is, how it is measured and predicted, how predictions are judged, and what each brief will ask of you.

> **Research goal:** a reproducible, potentially publishable contribution — research that could support a peer-reviewed paper, not a technical exercise. Publication is the ambition, not a requirement for passing.

Master 2 GENIOMHE-AI · October 2026 33 slides · read with the project briefs

RNA is information, machine and regulator at once

Carrier

mRNA carries coding information. Its untranslated regions, splicing and chemical modifications decide how much protein is made, and where.

Machine

Ribosomal RNA catalyses peptide-bond formation; spliceosomal snRNAs and ribozymes cut and join RNA. Here the structure is the function.

Regulator

Riboswitches, microRNAs, long non-coding RNAs and viral elements control gene expression through folding and base pairing.

RNA is now both a drug and a drug target: mRNA vaccines, antisense oligonucleotides, and small molecules that bind structured RNA.

The RNA zoo: what the briefs are actually about

Class	Typical size	What it does	Where it appears here
mRNA	0.5–10 kb	Carries the code; its UTRs regulate output	Modifications, language models
tRNA	76–90 nt	Adaptor in translation	Briefs 4 and 14: the most modified RNA
rRNA	120–5,000 nt	Catalyses peptide-bond formation	Large cryo-EM structures, ligand sites
snRNA, snoRNA	60–300 nt	Splicing; guides for 2′-O-me and Ψ	Modification maps, 3D benchmarks
microRNA, siRNA	22 nt mature	Silence target mRNAs	Precursor hairpins and bound complexes, not the mature strand
lncRNA, circRNA	200 nt and up	Scaffolds, decoys, regulation	Hard cases: long, weakly conserved
Riboswitches, ribozymes, aptamers	30–200 nt	Sense ligands, catalyse, bind	Briefs 2, 3, 15: the 3D workhorses
Viral elements	50–500 nt	IRES, frameshift and UTR signals	Brief 18

Short, structured, well-conserved RNAs are where prediction works and where almost every benchmark lives; long, weakly conserved RNAs are where it does not.

Four levels of structure, each with its own prediction problem

1 · PRIMARY

Sequence

A, C, G, U, plus more than 170 chemical modifications. Everything else is inferred from it.

2 · SECONDARY

Base pairs

Canonical A–U, G–C and G–U pairs form helices, loops, junctions and pseudoknots. Written in dot-bracket notation.

3 · TERTIARY

The 3D fold

Non-canonical pairs (Leontis–Westhof classes), stacking, A-minor motifs, coaxial helices and bound ions.

4 · COMPLEXES

With partners

RNA with proteins, other RNAs or small molecules: ribosomes, spliceosomes, riboswitch–ligand pairs.

Secondary structure can be inferred from sequence, thermodynamics, covariation and probing, though it stays uncertain in hard cases; the tertiary level and complexes are where prediction is hardest, and where briefs 1, 3, 4 and 17 live.

Secondary structure in one line: dot-bracket notation

A HAIRPIN

GGGAAAUCCC

(((....)))

Matching brackets are paired nucleotides: G1–C10, G2–C9, G3–C8. Dots are unpaired.

Element	Signature	Meaning
Stem (helix)	((((...))))	Consecutive pairs; the stable part of the fold
Hairpin loop	((....))	Unpaired run closing a stem
Bulge, internal loop	((.((..)).))	Unpaired nucleotides inside a stem
Multiloop	((..((..))..((..))..))	Several stems meeting at a junction
Pseudoknot	((..[[..))..]]	Crossing pairs; needs a second bracket type

Most thermodynamic algorithms exclude pseudoknots by design; several deep-learning predictors, SPOT-RNA and RibonanzaNet among them, do predict crossing pairs. Support depends on the predictor and on the output format: a downstream tool may silently drop pairs it cannot represent.

Beyond Watson–Crick: the tertiary alphabet

The helix

RNA helices are A-form: about eleven base pairs per turn, riboses in the C3′-endo pucker, a deep narrow major groove and a wide shallow minor groove.

That geometry is why proteins and small molecules usually read the minor groove, and why loops and junctions, not helices, carry specificity.

Pair geometry

Each base offers three edges: Watson–Crick, Hoogsteen and sugar. A pair is named by the two edges it uses plus a cis or trans orientation, which gives the twelve Leontis–Westhof families.

Non-canonical pairs are not rare: in a folded RNA they can be a third of all pairs.

Recurring modules

GNRA tetraloops and their receptors, kink-turns, A-minor contacts, ribose zippers, coaxially stacked helices, pseudoknots.

Annotated by FR3D, RNApdbee or barnaba, and catalogued in the RNA 3D Motif Atlas.

The INF metrics score exactly these interactions — a family of measures, with separate variants for base pairs and stacking and definitions that depend on the implementation. Blind assessments report these interactions as the weak point of current predictions, and brief 17 asks whether language models encode them at all.

Why RNA folds: the nearest-neighbour model

ADDING UP A HAIRPIN

GGGAAAUCCC

(((....)))

Two stacked pairs inside the stem are stabilising, roughly −2 to −3 kcal/mol each; closing a four-nucleotide loop costs roughly +5. The sum, a few kcal/mol below zero, is the structure's free energy.

Illustrative values only: each stack and loop energy depends on sequence and context. Parameters from the Turner set.

The energy model. Stability comes from stacking between neighbouring pairs rather than from pairs in isolation. Turner's parameters, measured by melting short duplexes, assign a free energy to every stack and a penalty to every loop, bulge and junction.

Two algorithms. Zuker's dynamic programming finds the minimum free energy structure in time proportional to the cube of the length. McCaskill's partition function instead sums over all structures, which yields base-pair probabilities, Boltzmann sampling and a measure of how certain the fold is.

The limits. The standard recursions exclude pseudoknots; the parameters know nothing of ions, proteins, modifications or the order in which the molecule was transcribed. Reported accuracy varies with the benchmark — by family, length, structural class and metric — so a single headline percentage says very little.

One sequence, several structures

It folds while it is made

Transcription is directional, so the 5′ end folds before the 3′ end exists. A kinetic trap formed early can outlive the structure that thermodynamics would prefer.

Conditions matter

Magnesium screens the backbone charge and lets helices pack; salt, temperature and crowding all move the equilibrium. CASP16 assessed ion and water placement around RNA for the first time.

Partners select states

Proteins, other RNAs and small molecules stabilise one conformation out of several. The structure in a crystal, in a test tube and in a cell need not be the same one.

So what do you report?

An ensemble, or a single structure with its conditions attached. Probing gives a population average; NMR and cryo-EM classes are experimental evidence of alternative states; molecular dynamics generates simulated ensembles that still need validating.

Brief 3 turns this into something measurable: when the ligand is declared, does a predictor return the state that matches it, or always the same one?

Experimental 3D structure is scarce; chemical probing is abundant

3D structures

X-ray crystallography, cryo-electron microscopy (now the workhorse for large RNAs) and NMR for small, dynamic RNAs. All deposited in the PDB.

RNA-only entries are a small share of the PDB — near 1% on a 2022 count, though the figure moves depending on whether you count entries, chains or any structure containing RNA — and they skew toward tRNAs, riboswitches and ribosomes. Benchmarks therefore hold tens to a few hundred targets.

Chemical probing

SHAPE reagents (2A3, 1M7) and DMS modify flexible, mostly unpaired nucleotides; mutational profiling and sequencing read the marks out.

The result is a per-nucleotide reactivity, not a list of pairs. Ribonanza measured about two million sequences this way; it is cheap and scales.

Consequence for your project: 3D evaluations are small-sample problems where statistics and leakage control decide credibility; probing data are large but indirect, so the mapping from reactivity to structure must be stated explicitly.

Reading RNA: what each technology actually measures

Method	What is read	What you get	Main caveat
Short-read RNA-seq	cDNA fragments of 50–300 nt	Expression, splice junctions	Isoforms stay ambiguous
Long-read cDNA (PacBio, ONT)	Full-length cDNA	Isoform structures	Amplification and length bias
Direct RNA nanopore (RNA004)	The RNA strand itself, as current	Reads of native RNA, completeness varying by experiment; modification signals	Calls depend on the basecaller model
Probing (SHAPE-MaP, DMS-MaPseq)	Mutations left at modified positions	Per-nucleotide reactivity	Reactivity is flexibility, not pairing
CLIP family (eCLIP, iCLIP)	Crosslinked protein–RNA sites	Binding sites along transcripts	Crosslinking and antibody bias
Proximity ligation (PARIS, SPLASH)	Ligated RNA duplexes	RNA–RNA contacts in cells	Sparse and noisy

Every row is an indirect measurement with its own biases. Half of the work in briefs 7 and 14 is stating exactly what the signal means before using it as a label.

Where the data live

Resource	What it holds	Used in briefs
PDB (RCSB)	Experimental 3D structures in mmCIF; ligand definitions in the Chemical Component Dictionary	1–6, 15–17
Rfam (15.1, January 2026)	4,227 RNA families: seed alignments, consensus secondary structures, covariance models, PDB mappings	9, 10, 15, 17, 18
RNAcentral	Tens of millions of non-coding RNA sequences; a major source for several language-model pretraining corpora	9, 10
MODOMICS, RMBase, REDIportal	Modification types, positions in tRNA and rRNA, transcriptome-wide sites, A-to-I editing	4, 14, 17
bpRNA, ArchiveII, bpRNA-new	Secondary-structure benchmark sets, with and without family-level splits	7, 10, 11
Ribonanza	Chemical mapping on about two million sequences, plus the RibonanzaNet model	7
RNA3DB	Curated PDB RNA chains, clustered by sequence and structure to reduce train/test redundancy	1, 9, 11, 16
HARIBOSS	RNA–small-molecule complexes from the PDB, curated for docking studies	2

The file formats you will actually touch

Format	Holds	Produced and read by
FASTA	Sequences	Everything
Stockholm	Alignments with family annotation, consensus structure included	Rfam, Infernal
A3M	Alignments in the shape most predictors expect, with no structure line	Co-folding and deep-learning predictors
Dot-bracket, CT, BPSEQ	Secondary structure	ViennaRNA, RNAstructure, bpRNA
PDB, mmCIF	3D coordinates	RCSB, every 3D tool
SMILES, SDF, CCD codes	Ligands and modified residues	RDKit, docking, co-folding inputs
BED, GTF	Coordinates and annotation	Modification sites, transcripts
SAM, BAM, CRAM	Read alignments	RNA-seq, nanopore pipelines
POD5	Raw nanopore signal	Basecalling, modification calling

Conventions that cause bugs. BED is zero-based and half-open; GTF and PDB numbering are one-based; PDB residue numbers may skip, repeat or carry insertion codes. A residue's number is not its index.

PREDICTION

Finding relatives: why RNA needs more than BLAST

Sequence search runs out

BLASTN is reliable while relatives stay close in sequence. Structured RNAs do the opposite: they conserve the fold and let the sequence drift, so two riboswitches of the same family can share an architecture and almost no identity.

Covariance models

Infernal scores sequence and secondary structure at the same time, so a compensatory mutation counts as evidence rather than as a mismatch. This is how Rfam defines its families, and how remote homologs are found at all.

Scores

Bit scores and E-values, plus the per-family gathering, noise and trusted cut-offs that Rfam curates by hand.

Clans

Families too similar to separate are grouped, so that one locus is not counted twice as two different RNAs.

Depth

What helps a predictor is the effective number of sequences, not the raw count: a thousand near-identical hits carry little information.

Evolution reveals structure: covariation

THREE HOMOLOGS, ONE STEM

GGCGAAAACGCC

GACGAAAACGUC

GUCGAAAACGAC

((((....))))

Positions 2 and 11 change together (G–C, A–U, U–A) and stay paired: a compensatory mutation.

A base pair that matters for function survives mutation only if both partners change. Across an alignment of homologs, columns that covary this way mark the helices.

Tools. Alignment: MAFFT, LocARNA. Family models: Infernal covariance models, the engine behind Rfam. Significance of covariation: R-scape. Consensus folding: RNAalifold.

Why 3D predictors care. Deep alignments drive the RNA-specific predictors such as RhoFold+ and trRosettaRNA; rMSA and RNAcmap3 build them from RNAcentral and NCBI nucleotide data.

Support is model- and version-specific. AlphaFold 3 and selected Protenix versions document RNA alignment input; Boltz-2's documented custom-alignment field is protein-only; for Chai-1, verify the workflow in the exact release you use. Being able to build an alignment is not evidence that a predictor consumes it.

Three ways to predict secondary structure

Thermodynamic

A nearest-neighbour energy model; dynamic programming finds the minimum free energy structure and the partition function yields base-pair probabilities.

ViennaRNA, RNAstructure, LinearPartition. No training data; pseudoknots usually excluded.

Probabilistic

Stochastic grammars whose parameters are learned from known structures or from probing data; still output pair probabilities.

CONTRAfold, EternaFold. A strong middle ground between physics and learning.

Deep learning

Neural networks trained on known structures. Strong within the families they have seen; across new families results vary by model and benchmark, and several studies report a marked drop.

SPOT-RNA, UFold, MXfold2, RibonanzaNet. Some handle pseudoknots.

Rule of thumb: include a task-appropriate baseline, often the thermodynamic model, and read any learned score together with its test split. "Helps on a subset of targets" is a legitimate finding.

3D prediction: three generations of methods

Generation	Examples	Inputs	Strengths and limits
Physics and fragment assembly	FARFAR2 (Rosetta), SimRNA, RNAComposer	Sequence, a secondary structure, sampling	Interpretable, usable for refinement; slow, and only as good as the 2D input
RNA-specific deep learning	trRosettaRNA, RhoFold+, DRfold2, DeepFoldRNA	Sequence, alignment, language-model embeddings	Fast; accuracy tracks alignment depth and family coverage
All-atom co-folding	AlphaFold 3, Boltz-2, Chai-1, Protenix, OpenFold3, RNAPro	Sequence, alignment, templates, ligands, modifications, partner chains	Handles complexes and ligands, and reports confidence scores; GPU-bound, and RNA alignment support varies by model and release

Confidence outputs differ by predictor: some report a per-residue estimate, some a predicted error matrix or an interface score, some none at all. They are not comparable across tools, and a confidence value is not demonstrated accuracy — which is what brief 6 measures.

PREDICTION

Where blind tests say we stand

Reliable

Helices and most of the secondary structure; global folds when close homologs exist in the PDB.

Unreliable

Non-canonical pairs, coaxial stacking, chain entanglements: the regions that carry function, such as active sites.

Open

RNA–RNA multimers, RNA–ligand sites, ensembles of alternative conformations and solvation were added as CASP16 targets. CASP17 closed its prediction season in September 2026 and assessment is still under way, so treat it as an open experiment rather than a source of results.

Caution

Accuracy falls with distance from training structures, so a benchmark reports performance against similarity; a decline is an association, not proof of memorization.

Scoring a 3D model

Metric	What it measures	How to read it
RMSD	Root-mean-square deviation of aligned atoms after superposition	Lower is better; grows with length; one bad region dominates
TM-score	Length-normalized structural similarity, 0 to 1	Higher is more similar; 0.45–0.5 is a rough heuristic, not a universal rule
lDDT	Local distance agreement, no superposition, 0 to 1	Per-nucleotide quality; robust to hinge motions
INF	Fidelity of the interaction network: pairs and stacking	RNA-specific; split into canonical, non-canonical, stacking
Clash score	Steric overlaps per 1,000 atoms	Physical plausibility, not accuracy
pLDDT, ipTM, where supplied	The predictor's own confidence estimate	Needs no reference; not comparable across tools; calibrate before trusting

US-align computes TM-score; RNAdvisor bundles the rest. Without a reference structure, model quality assessors (RNArank, lociPARSE) estimate quality from the model alone.

EVALUATION

How to benchmark without fooling yourself

Temporal split

Test on structures released after the model's documented cutoff. It lowers the risk of overlap but does not prove a target was unseen.

Similarity stratification

Bin test cases by sequence identity and structural similarity to the training data; report the curve, not one average.

Family-level split

Hold out whole Rfam families. Within-family accuracy says nothing about a family the model never saw.

Baselines and uncertainty

Run the simple baseline (ViennaRNA, docking), bootstrap confidence intervals over targets, and freeze the protocol before looking at results.

Why this matters for your project: a high score on familiar RNA families may not survive on new families, or on structures released after training.

Four conventions behind every number you will read

The split is the experiment

A random split over RNA families measures interpolation inside families the model has seen. Family and temporal splits measure what you actually want to know. Report the split before the score.

Rare labels break metrics

Most candidate pairs or sites are negatives, and with that imbalance AUROC flatters almost anything. AUPRC and the Matthews correlation move when the rare class is wrong; prefer them.

Scores, not just labels

A model that outputs a probability can be ranked, thresholded and calibrated. Calibration asks a separate question from accuracy: are the cases given 0.8 right about 80% of the time? Brief 7 measures exactly that.

Three ways to use a pretrained model

Zero-shot: score with the model as it is. Fine-tuned: train a head on your task. Probed: freeze it and ask what a linear classifier can read from its embeddings. Briefs 10 and 17 use the last two.

EVALUATION

Reporting a benchmark so that a reader can trust it

The unit is the target

Forty targets give forty target-level observations, not four thousand nucleotides. Resample whole targets, and account for targets that share a family or cluster.

Compare in pairs

Run every tool on the same targets and test the paired differences, for example with a Wilcoxon signed-rank test. Means over two different target sets cannot be compared at all.

Intervals over verdicts

Report the median difference with a confidence interval. On a few dozen targets, small gaps in TM-score are noise, and saying so is a result rather than a failure.

Count your comparisons

Several tools times several metrics times several bins is a lot of tests. Correct for them, say how many were run, and fix the seeds so that a rerun gives the same table.

And remember what a score is measured against: a benchmark label is an operational definition, not biological truth. Say what the reference measurement represents, what uncertainty it carries and which claims it can support.

MACHINE LEARNING

RNA language models: embeddings from millions of sequences

How they work

A transformer is trained to recover masked nucleotides across RNAcentral-scale corpora. It returns a vector per nucleotide that downstream models fine-tune, score zero-shot, or probe.

Examples: RNA-FM, RiNALMo, ERNIE-RNA, UTR-LM, SpliceBERT.

How they are judged

Benchmarks such as BEACON (13 tasks) and RNA-Scope compare models on structure, interaction and function tasks; a 2026 study in Briefings in Bioinformatics scored 21 models zero-shot.

RiNALMo's authors report that fine-tuning generalizes across held-out families on secondary structure: a result on that benchmark, not a guarantee for any unseen family.

Caveats

Plain CNN and LSTM baselines beat language models on several benchmark tasks. Pretraining corpora can overlap test sets. What an embedding encodes beyond pairing is an open question.

Briefs 10 and 17 ask exactly the two open questions: how much overlap inflates the benchmarks, and what the embeddings know about non-canonical structure.

EPITRANSCRIPTOME

More than 170 modified nucleotides, five that matter most

Modification	Where it occurs	Detection or evidence source
m6A	mRNA (DRACH motifs), rRNA, viral RNA	Antibody capture (MeRIP), chemical conversion (GLORI, eTAM-seq), nanopore direct RNA
Pseudouridine (Ψ)	tRNA, rRNA, snRNA, mRNA	Bisulfite-based deletion signatures (BID-seq, PRAISE), nanopore
m5C	tRNA, rRNA, some mRNA	Bisulfite sequencing, nanopore
Inosine (A-to-I editing)	mRNA, double-stranded regions	RNA-seq mismatches against the genome, checked against DNA to exclude variants; nanopore; REDIportal curates sites
2′-O-methylation	rRNA, snRNA, mRNA caps	Nm-seq, RibOxi-seq, nanopore

A tRNA carries 5 to 15 modifications; almost every structure predictor ignores them. Each detection method has its own biases and coverage, so no single map is ground truth.

RNA as a drug target: riboswitches, aptamers, small molecules

Riboswitches

An aptamer domain binds a metabolite; the expression platform then switches between ON and OFF structures. Ligand-free and ligand-bound structures exist for many classes, which makes them a natural test of conformational switching.

Small molecules

Risdiplam, an approved drug, modulates SMN2 splicing by binding pre-mRNA. RNA pockets are shallow and flexible, and curated RNA–ligand benchmarks stay small next to protein–ligand ones; how small depends on the inclusion criteria and redundancy filters.

Methods

Docking (rDock, AnnapuRNA) places a ligand into a known structure; co-folding (AlphaFold 3, Boltz-2, Chai-1) predicts RNA and ligand together. The metric is ligand RMSD after superposing the pocket.

Open question behind briefs 2, 3 and 18: co-folding has not yet shown consistent success on RNA–ligand complexes, and how it compares with RNA-aware docking on independent targets carrying novel ligands or pockets is still open.

Inverse folding: from a target structure to a sequence

2D design

Find a sequence whose predicted secondary structure matches the target: RNAinverse, antaRNA and newer learned designers. The Eterna100 benchmark showed that difficulty depends on which folding engine judges the design.

3D design

Geometric deep learning on backbones: gRNAde, RiboDiffusion, RNA-FrameFlow. Judged by sequence recovery and by self-consistency: does the designed sequence refold into the target?



gRNAde already reports this loop with RhoFold as judge; brief 16 asks what independent judges do to its designability estimates, and how often the judges disagree.

PRACTICE

The working toolbox

Task	Tools	Compute
Secondary structure	ViennaRNA, LinearPartition, CONTRAfold, EternaFold, RNAstructure	CPU
Alignments and families	Infernal, MMseqs2, MAFFT, LocARNA, R-scape, rMSA	CPU; large databases need disk
3D prediction	Boltz-2, Chai-1, Protenix, OpenFold3, RNAPro	GPU
3D comparison and annotation	US-align, RNAdvisor, barnaba, RNApdbee, MolProbity	CPU
Ligands	RDKit, rDock, AutoDock Vina, AnnapuRNA	CPU
Language models	RNA-FM, RiNALMo, ERNIE-RNA (inference only)	GPU; grows with model and batch size
Reproducibility	conda or Docker, pinned versions, run logs, a results table that regenerates from scripts	Discipline

Half of the briefs never need a GPU. For the other half, confirm access in week 1 and measure one real run before budgeting: requirements depend on the tool, the model version and the input, and co-folding scales with targets × models × seeds.

PRACTICE

How the semester runs: 16 weeks, four gates



Roles

Harness owner, data owner, one tool runner per predictor, writer. Small teams rotate; large teams split work packages under one protocol and one harness.

Protocol

Tools, versions, data releases, metrics and hypotheses written down before any result is seen; later changes logged with a reason.

Minimum viable version

Every brief sorts its work into required (the MVP, week 8), one main extension and stretch. Stretch work is never an implicit requirement.

The 16 briefs at a glance (1 of 2)

Brief	Knowledge it rests on	Main skill	Difficulty · compute
1 Temporal test of co-folding models	3D prediction, metrics, splits	Pipelines, statistics	4 · GPU heavy
2 Docking vs co-folding, RNA–ligand	Ligands, pockets, metrics	Chemoinformatics	4 · GPU
3 Riboswitch switching	Riboswitches, tertiary structure	Structural biology	3 · GPU light
4 Modified nucleotides as input	Modifications, 3D metrics	Curation	2 · GPU light
5 Stereochemistry and refinement	Geometry, validation	Structural validation	2 · CPU
6 Model selection	Metrics, confidence	Statistics	3 · CPU
7 Calibration of pair probabilities	Thermodynamics, probing	Statistics	2 · CPU
9 MSA ablation	Covariation, alignment-consuming predictors	Pipelines	3 · CPU then GPU

The 16 briefs at a glance (2 of 2)

Brief	Knowledge it rests on	Main skill	Difficulty · compute
10 Leakage audit of LM benchmarks	Language models, families	Data auditing	2 · CPU
11 Replication study	Depends on the paper	Reproducibility	3 · varies
12 Systematic review	Evaluation practice	Reading, annotation	2 · none
14 Modification evidence atlas	Modifications, detection methods	Data curation	3 · none
15 Rfam structural dark matter	Families, 3D prediction	Pipelines	3 · GPU
16 Independent evaluation of inverse folding	Inverse folding, metrics	ML pipelines	4 · GPU heavy
17 Probing LM embeddings	Language models, tertiary motifs	Machine learning	3 · GPU light
18 Viral RNA elements	Covariation, 2D and 3D, ligands	Structural analysis	3 · GPU light

CHOOSING

Four questions before you pick

What do you want to learn?

Structural biology: 3, 4, 5, 15, 18. Machine learning: 10, 16, 17. Data engineering: 1, 9, 14. How science is evaluated: 11, 12.

What compute can you reach?

No GPU at all: 5, 6, 7, 10, 12, 14. A few GPU hours: 3, 4, 17, 18. Guaranteed GPU budget: 1, 2, 9, 15, 16.

How big is the team?

There is no upper limit. Recommended: two or three for 4, 5, 7, 11, 17; four to six for 1, 2, 12, 16. Larger teams split work packages under one protocol.

How much risk do you want?

Low, a clear null result still counts: 4, 5, 7, 10. Higher payoff with more moving parts: 1, 2, 16. Every brief names a fallback.

REFERENCE

Twenty terms you will meet in the briefs

MFE — the single lowest-free-energy structure

Partition function — the weighted sum over all structures

Base-pair probability — how often a pair occurs in that ensemble

Pseudoknot — crossing pairs, excluded by most 2D algorithms

Dot-bracket — the one-line notation for a secondary structure

Leontis–Westhof — the twelve families of base-pair geometry

Coaxial stacking — two helices stacking end to end at a junction

Covariation — alignment columns that mutate together

Covariance model — a profile scoring sequence and structure at once

Neff — the effective number of sequences in an alignment

Template — a known structure handed to a predictor

Co-folding — RNA and its partners predicted together, all atoms

pLDDT — a per-residue confidence score, where a model supplies one

TM-score — length-normalized similarity, 0 to 1; cut-offs are heuristics

lDDT — local distance agreement, computed without superposition

INF — fidelity of the pair and stacking network

Inverse folding — designing a sequence for a target structure

Self-consistency — refolding a design to see whether it returns

Temporal split — testing on structures released after a cutoff date

Leakage — test data the model effectively saw already

REFERENCE

Where to go deeper in week 1

Tools and manuals

ViennaRNA — the energy model and its algorithms

Infernal — covariance models; the user guide is the best primer

Rfam 15.1 and RNAcentral — families and sequences

RNAdvisor — the 3D metrics in one place

Blind assessments

CASP16 — RNA targets and assessors' reports

CASP17 — under assessment, results pending

RNA-Puzzles Round V — 23 targets, 18 groups

RNA3DB — redundancy-aware test sets

Learning and data

Szikszai et al., 2026 — the generalization gap

CHANRG — fair splits versus the leaderboard

BEACON — 13 tasks, strong simple baselines

Ribonanza and gRNAde — probing at scale, and 3D design

Tool capabilities change between releases: record the version you tested and the date you read its documentation. Full citations, with versions and check dates, are in the reference section of the briefs document.

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Read the briefs, form a team, rank three

1. Read the 16 project briefs: question, data, tools, metrics, week-by-week plan, risks.
2. Form a team — the sizes in the overview are recommendations, not limits — and rank three briefs, with a sentence on why each fits you.
3. Send your ranking by the end of week 1; GPU access questions go to guillaume.postic@universite-paris-saclay.fr.
4. Week 1: environment installed, roles assigned, three test cases run end to end.